

Seeing What Few-Shot Learners See: Contrastive Cross-Class Attribution for Explainability

Lingfeng Chen^{ID}, Panhe Hu^{ID}, and Zhen Liu^{ID}

Abstract—Few-shot learning (FSL) enables deep learning models to generalize to unseen categories with minimal labeled data, making it crucial for data-constrained domains such as healthcare. However, existing FSL models often lack explainability, obscuring the reasoning behind their predictions and limiting trust in their deployment. In this work, we introduce contrastive cross-class attribution (C3A) to enhance explainability in FSL. Specifically, we present C3A for explaining query predictions and its variant (C3A-E) for episodic learning. Unlike prior explainability methods designed for specific architectures, our approach is classifier-agnostic, leveraging the backbone network to ensure broad applicability across diverse FSL paradigms. In addition, we propose contrastive local descriptors that integrate naive-Bayes nearest-neighbor (NBNN) and Fisher score-based attribution to preserve fine-grained details and enhance inter-class discrimination. Finally, we comprehensively evaluate our methods across four FSL benchmarks, demonstrating consistent improvements over state-of-the-art approaches, with absolute gains of 19.72% in insertion-area-under-curve (iAUC) and 25.88% in deletion-area-above-curve (dAAC). Our posthoc design ensures seamless integration with various architectures, including Conv64F, ResNet12, ResNet18, and ViT. The experimental results confirm that C3A and its variant (C3A-E) provide more stable and interpretable explanations compared to existing FSL explainability methods, offering deeper insights into the episodic learning process. We have made our code publicly available at <https://github.com/MountainChenCad/C3A>

Index Terms—Contrastive learning, explainable artificial intelligence, feature attribution, few-shot learning (FSL), neural networks.

I. INTRODUCTION

MODERN deep learning models rely on large-scale labeled datasets to achieve high performance [1], [2]. However, in critical domains such as healthcare (e.g. MRI scans), data acquisition is often constrained by privacy regulations, ethical concerns, and technical limitations [3], [4], [5], [6], [7]. This scarcity of labeled data poses a significant challenge to the deployment of traditional deep learning models in real-world applications. Consequently, models must learn

effectively from a limited number of examples, a paradigm known as few-shot learning (FSL). FSL enables models to generalize to unseen categories with limited data, offering a viable approach for data-constrained scenarios [8], [9].

The standard evaluation framework for FSL employs the N -way K -shot (i.e., episodic) paradigm, where N represents the number of classes and K denotes the quantity of labeled exemplars per class. During the training phase, models are exposed to a support set $\mathcal{S}$ containing $N \times K$ samples and subsequently evaluated on a distinct query set $\mathcal{Q}$ [8]. However, a major limitation of existing FSL systems is their lack of explainability, which obscures the reasoning behind their predictions [7], [10], [11], [12], [13], [14], [15], [16], [17]. This opacity is particularly problematic in high-stakes applications. For instance, in few-shot medical image diagnosis, clinicians must verify that a model's prediction is based on relevant pathological features rather than confounding artifacts. Similarly, in fine-grained tasks (e.g., distinguishing bird species), it is essential to verify that the model attends to subtle, unique traits rather than background context. Therefore, comprehending *how an FSL model generalizes to unseen categories by correctly matching images* is crucial for optimizing model performance, mitigating potential risks, and fostering user trust [7], [14], [18], [19], [20], [21], [22].

Explainable AI (XAI) techniques are essential for understanding deep learning models [7], [20], [21], [23], [24]. However, most XAI research has focused on conventional data-driven models, with limited attention to FSL systems. Traditional XAI methods like SHapley Additive exPlanation (SHAP) often require access to a large portion of the training set, which is not feasible in FSL settings where only a small support set is available [7].

To our knowledge, the first attempt to introduce explainability in FSL was the *Siamese Networks Explainer* [SINEX(C)], designed to interpret the episodic learning process of Siamese networks in acoustic data tasks [16]. However, SINEX(C) is inherently tailored to Siamese architectures, limiting its generalizability to other FSL paradigms and restricting its applicability beyond episodic explanations. In addition, its local interpretable model-agnostic explanation (LIME)-like superpixel segmentation produces inconsistent explanations, further compromising its reliability (See further analysis in Section S.B1).

More recently, Samuel and Gregory [25] introduced *Proto-ShotXAI*, an FSL-compatible explainability framework based on Prototypical Networks (ProtoNet) [26]. It explains FSL

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queries using image-level feature attribution maps. However, ProtoShotXAI relies on classification layer weights, which limits its applicability to FSL frameworks that do not incorporate such weights (Often the case). To ensure a fair comparison, we modified ProtoShotXAI into a variant, named ProtoShot*, removing the dependence on classification layer weights. Experimental results indicate that while ProtoShot* performs similarly to ProtoShotXAI in traditional settings, it is less effective in explaining the FSL backbone, as shown in Fig. 1 (Additional comparative results between ProtoShotXAI and ProtoShot* are in Section S.B2). Fig. 1 reveals a critical insight: while ImageNet-pretrained backbones (e.g., ResNet50) contain rich, general-purpose features that yield clear maps, FSL-trained backbones (e.g., Conv64F and ResNet12) tend to compress representations to optimize metric distance. This compression leads to “over-smoothed” and unstable attribution maps where standard XAI methods fail to distinguish foreground from background. This highlights a fundamental gap: existing methods are insufficient for recovering discriminative details from the highly entangled feature spaces typical of FSL models.

Versatility: As noted in current research [27], [28], [29], [30], [31], although contemporary FSL methods employ diverse architectures, they rely heavily on the feature extraction capabilities of backbone networks, especially in low-domain-shift scenarios. To enhance the flexibility of explainability in FSL, we propose contrastive cross-class attribution (C3A), a classifier-agnostic framework that focuses on explaining the backbone rather than the task-specific head, ensuring broad applicability across different FSL paradigms.

Contrastive Local Descriptor: Image-level representations inherently discard discriminative details, leading to the over-smoothing issue identified above. To address this, we propose contrastive local descriptors that leverage the strengths of naive-Bayes nearest-neighbor (NBNN) and the *Fisher score* for improved feature attribution.

Specifically, we introduce an intraclass local descriptor to mitigate over-smoothing, drawing inspiration from NBNN’s success in few-shot classification. We argue that traditional image-level feature-based measures are the primary cause of over-smoothing and that an image-to-class approach is necessary [32], [33], [34], [35]. Crucially, while contrastive principles are widely used in *metric learning* to train classifiers, we utilize them here for *posthoc attribution*. Our core innovation is an image-to-multiclass contrastive mechanism that calculates attribution not just by similarity to the target class, but by explicitly contrasting it against competing classes. This ensures that our framework not only captures class-specific information but also aggressively filters out nondiscriminative background noise, enhancing feature discrimination across multiple categories. The proposed contrastive local descriptor has two key insights as follows.

1) **Preserving Local Information:** Instead of treating an image as a single representation, we model it as a collection of local descriptors, preserving fine-grained details within the same class in the support set.

2) **Enhancing Discriminative Power:** A feature’s strong similarity to one class does not guarantee its uniqueness

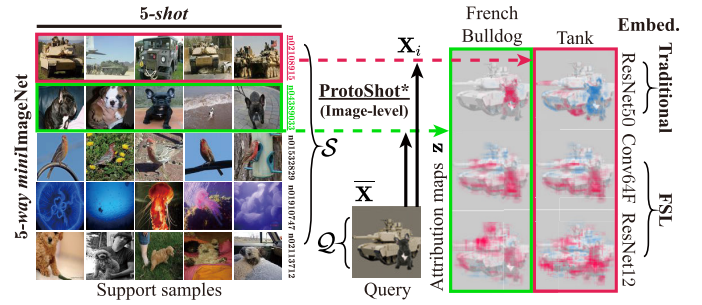


Fig. 1. ProtoShotXAI’s key limitation emerges in FSL-trained backbones despite strong ImageNet-pretrained ResNet50 performance. Left: 5-way 5-shot (i.e., five samples each for n02108915, n04389023, n01532829, n01910747, and n02113712) classification task example for *miniImageNet*, only 1-way of samples are utilized for each explanation. Middle: Query sample. Right: Inconsistent attribution maps for Tank versus French Bulldog, revealing confusing patterns in FSL backbones (Conv64F and ResNet12) versus clear distinctions in ImageNet-pretrained ResNet50.

if it also closely resembles other classes. By incorporating contrastive interclass information, our method ensures a more precise and robust attribution measure.

Comprehensiveness: Existing methods, SINEX(C), focused on explaining the episodic learning in FSL, and ProtoShotXAI focuses on interpreting FSL queries (traditional XAI). None of them can directly provide both. However, in FSL, understanding how models adapt to novel categories with minimal examples in an episode is equally crucial.

To offer a comprehensive explanation of FSL models, we introduce C3A for interpreting FSL queries and its variant, C3A-E, for episodic learning—an explainability framework that extends beyond individual query predictions. Specifically, C3A-E elucidates how an FSL model processes a query instance $\bar{\mathbf{X}}$ in relation to a support set $\mathcal{S}$, even when provided with only a few examples per class.

Our Primary Contributions:

- 1) **Flexible Classifier-Agnostic Framework:** Our proposed model, C3A, focuses on explaining the backbone rather than the task-specific head, ensuring broad applicability across diverse FSL paradigms.
- 2) **Contrastive Local Descriptors for Robust Feature Attribution:** By leveraging local descriptors and an interclass Fisher discriminant measure, our method enhances explainability, specifically addressing the over-smoothing and instability issues prevalent in FSL backbones.
- 3) **Comprehensive Explainability for Episodic Learning:** Our framework elucidates how FSL models generalize with limited examples. It introduces C3A for explaining queries and its variant, C3A-E, for episodes, extending beyond traditional single-instance explanations.
- 4) **Empirical Comparison With Existing FSL Explainability Methods:** We evaluated our approach against state-of-the-art baselines across multiple datasets and FSL backbones, demonstrating superior performance and versatility across diverse FSL settings.

II. RELATED WORK

A. Few-Shot Learning

Our contribution serves as a framework for posthoc explanation on FSL backbones. Current important FSL methods can be roughly divided as follows.

Meta-Learning Based Methods: These methods address FSL by training meta-learners to “learn how to learn” [36], [37], [38], [39]. Finn et al. [36] introduced Model-Agnostic Meta-Learning (MAML), optimizing model initialization for rapid task adaptation. Meta-SGD [37] extends MAML by optimizing learning rates and updating directions. Ravi and Larochelle [38] proposed Meta-LSTM, using LSTM to dynamically update model parameters via an external memory module. MM-Net [39] predicts embedding network parameters through memory slots for unlabeled images. Recently, meta-learning has evolved toward highly adaptive and modular frameworks to handle noise and instability. For instance, Agrawal et al. introduced adaptive FSL (AFSL) [40], which integrates dynamic stability modules and noise-adaptive resilience mechanisms to mitigate performance variance. While these intrinsic methods enhance model robustness during training, they often require complex architectural modifications and extensive retraining costs. In contrast, C3A is a posthoc framework that extracts explanations from existing models without altering their training pipeline.

Metric Learning-Based Methods: These methods focus on feature comparison [26], [32], [41], [42]. MatchingNet [42] introduces an attention-based mechanism to compare query samples with support sets, enabling end-to-end FSL. Siamese network (SN) [41] learns a similarity metric between pairs of inputs, making it effective for tasks such as verification and matching. ProtoNet [26] leverages the concept of prototypes, representing each class by its mean feature vector, and classifies queries based on their distance to these image-level prototypes (i.e., one for each sample). Subsequently, Li et al. [32] proposed a deep nearest neighbor neural network (DN4) based on NBNN that employs deep local descriptors and image-to-class metrics, significantly improving few-shot classification. However, existing meta-learning and metric-based approaches often lack transparency in explaining how backbone networks establish discriminative feature representations. Crucially, it is important to distinguish our work from these standard metric learning approaches. While they utilize contrastive similarity measures globally or locally for *classification*, C3A repurposes these principles for *attribution*. We do not claim to invent the contrastive similarity principle, but rather to novelly apply it as a post hoc mechanism to resolve the over-smoothing and feature entanglement issues in FSL explainability.

Causal-Based Methods: A simplified causal perspective has recently emerged to address the “spurious correlation” problem in FSL. Lin et al. [43] theoretically reframed metric learning as a form of Front-Door Adjustment, suggesting that a good feature extractor acts as a mediator that blocks confounders. Following this, specialized causal architectures have been proposed: Wen et al. [44] utilized causal intervention for prompt tuning to improve out-of-distribution generalization, and recent

works in SAR recognition [45] employed structural causal models (SCMs) to disentangle imaging conditions from target features. While theoretically grounded, these methods typically require constructing explicit causal graphs or performing heavy interventional training [46]. C3A achieves mitigating the influence of noncausal background noise through a computationally efficient contrastive metric, avoiding the need for complex SCM construction or retraining.

Our approach extends the work of Samuel and Gregory [25], who pioneered the use of FSL architectures for explainability. Although meta-learning methods often involve complex architectures, high computational costs, and training difficulties [47], metric learning based approaches offer a more efficient alternative. Although Lake et al. [48] introduced ProtoShotXAI based on ProtoNet, their evaluation was limited to simpler datasets such as Omniglot. Our experiments across four more challenging FSL benchmarks revealed significant limitations in ProtoShotXAI’s effectiveness with various FSL backbones. While C3A similarly leverages FSL architecture for explainability, we developed a more discriminative image-to-multiclass attribution measure that demonstrates superior performance in diverse scenarios.

B. Explainable AI

In this study, we focus on explaining feature attributions through FSL backbones, which inherently fall in close-box model explanation (i.e., “*posthoc*”). Hence, we briefly review two main branches of this direction as follows.

Traditional Posthoc XAI Methods: In this section, we introduce several well-established posthoc explanation methods, which includes: gradient-weighted class activation mapping (Grad-CAM) [49], which locates key regions through gradient-weighted class activation maps; LIMEs [50], which generates explanations through local linear approximations; SHAP [51], which calculates feature contributions based on game theory; randomized input sampling for explanation (RISE) [52], which generates saliency maps through random masking; Saliency maps [53], which use gradient information to generate feature attribution maps; and Anchors [54], which generate high-precision rules to explain model decisions. However, the direct application of conventional XAI methodologies to FSL paradigms encounters fundamental compatibility barriers [7]. First, methods like Grad-CAM and saliency maps rely heavily on gradient propagation through deep networks, which becomes unstable when applied to meta-learning models with episodic training and rapid adaptation mechanisms [49], [53], [55], [56]. Second, LIME and SHAP assume local linearity in feature space, an assumption that frequently breaks down in low-data regimes where decision boundaries are less constrained [7], [50], [51], [57], [58]. Finally, these post hoc explanation frameworks were designed for models trained on abundant data, and their reliable output often depends on access to large training data (e.g., SHAP), which is not applicable in FSL [7], [58], [59]. This mismatch often produces explanations that overemphasize artifact features of support sets while overlooking critical cross-episode meta-features [55], [58]. To the best of our knowledge, only Grad-CAM has been tested in FSL [16], and currently the effectiveness of

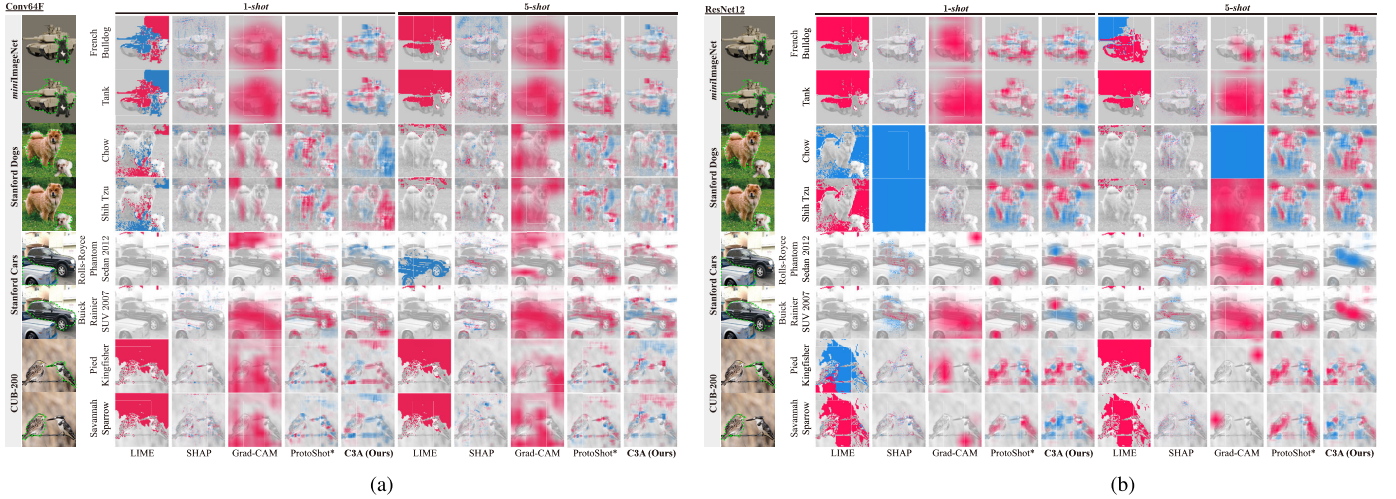


Fig. 2. Qualitative comparison of attribution maps from five XAI methods (LIME, Grad-SHAP, Grad-CAM, ProtoShot*, and ours C3A) trained separately on each dataset in *1-shot* and *5-shot* conditions. (a) Performance on Conv64F and (b) ResNet12. The query images span: Tank and French Bulldog (*miniImageNet*), Chow and Shih Tzu (Stanford Dogs), Rolls-Royce Phantom Sedan 2012 and Buick Rainier SUV 2007 (Stanford Cars), and Savannah Sparrow & Pied Kingfisher (CUB-200) (Enclosed by green dashed lines). Throughout both backbones, C3A demonstrates superior spatial specificity, with positive (red) and negative (blue) attributions precisely localized to class-discriminative regions. In contrast, traditional methods produce unstable results and ProtoShot* suffers from over-smoothing, particularly on ResNet12.

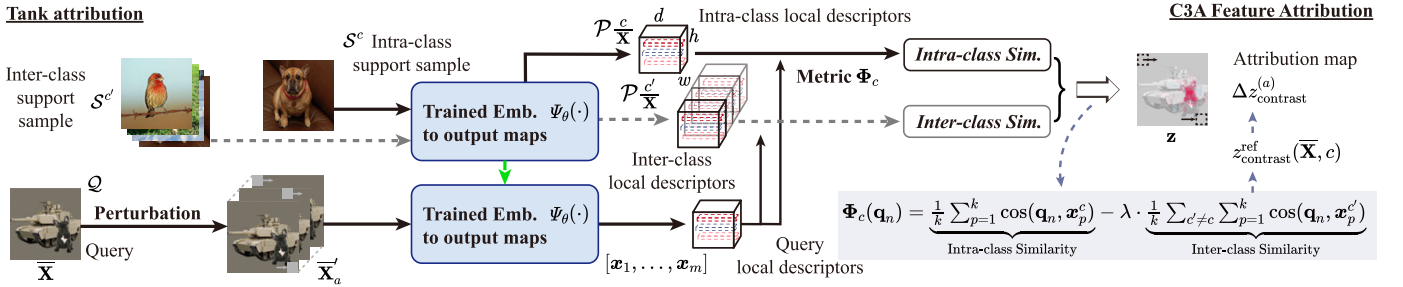


Fig. 3. Illustration of the proposed (C3A) for explaining a FSL backbone in *5-way 1-shot* scenarios. Top: Trained FSL model with task-specific explanation data. The C3A architecture contains three core branches: 1) target class support samples S^c (Tank); 2) contrast class samples $S^{c'}$ (French Bulldog, Wolf, etc.); and 3) query sample $\bar{X}$ perturbations. Local descriptor pools $\mathcal{P}_{\bar{X}}^c$ (Target) and $\mathcal{P}_{\bar{X}}^{c'}$ (Contrast) derive from backbone features of respective support sets. For each perturbed query image $\bar{X}_q'$, intra/interclass similarities are computed via NBN between query features and both descriptor pools, generating pixel-level attribution scores. Process repeats across all classes to produce positive/negative attributions with respect to class c .

methods like LIME and SHAP in FSL settings remains less explored.

Prior work on few-shot XAI methods. Recent efforts have introduced XAI techniques specifically designed for FSL contexts. SINEX [16] pioneered this direction by proposing an explanation method for SNs [41] on acoustic benchmarks. It segments acoustic images into superpixels, perturbs each segment, and quantifies its contribution based on the resulting similarity changes. ProtoShotXAI [25] extends ProtoNet's [26] approach by measuring how patch perturbations affect similarity scores to evaluate local feature attribution. More recently, Jo et al. [60] proposed template matching and regression (TMR) to preserve spatial structures by avoiding global pooling, and Donnelly et al. [61] introduced ProtoRSet to allow interactive debugging of prototype networks via Rashomon sets. While TMR offers structural interpretability, it is architecturally constrained to template-based detection tasks. Similarly, ProtoRSet requires human-in-the-loop intervention to select valid models. In contrast, C3A is fully automated and model-agnostic, applicable to both CNNs and ViTs without

architectural constraints. Despite its purported applicability to FSL, ProtoShotXAI has primarily been validated only on the relatively simple Omniglot benchmark. Our analysis reveals that its patch perturbation and image-level approach prove suboptimal for FSL contexts. While these methods represent progress, they exhibit significant limitations: SINEX(C) is constrained to SNs and heavily dependent on the model's similarity metric (See Section S.B1), while ProtoShotXAI suffers from instability due to image-level over-smoothing on FSL backbones [As shown in Fig. 1 and experimental results in Fig. 2(a) and (b)] and requires access to classification weights. Consequently, developing robust and generalizable XAI methods for FSL remains an important open challenge.

III. METHOD

A. Problem Formulation

In this section, the general model of the few-shot episodic image classification is introduced. For each task $\mathcal{T}$, consider an image sample $\mathbf{X} \in \mathbb{R}^{m \times q}$, where m, q denote the width and

length of the image, respectively. Then, a N -way K -shot task $\mathcal{T}$ can be expressed as follows:

$$\mathcal{T} = \{(\mathbf{X}_1, l_1), \dots, (\mathbf{X}_C, l_C)\}, \{\bar{\mathbf{X}}_1, \dots, \bar{\mathbf{X}}_t\} \quad (1)$$

where $l_i \in \{1, \dots, N\}$ is the label of $\mathbf{X}_i$ and footnote $i \in \{1, 2, \dots, C\}$. $\{\bar{\mathbf{X}}_1, \dots, \bar{\mathbf{X}}_t\}$ represents t samples as query set $\mathcal{Q}$ and $\{(\mathbf{X}_1, l_1), \dots, (\mathbf{X}_C, l_C)\}$ stands for K -shot support set $\mathcal{S}$ ($C = K \cdot N$). N means the number of classes the samples in $\mathcal{T}$ can be divided into. A FSL model is a deep learning model $f_\Theta(\mathcal{T}) = p(Y|\mathcal{T})$ determined by trainable parameters Θ (Training procedures see [8]) that takes as input a support set $\mathcal{S}$ containing images $\mathbf{X}_i$ and their corresponding class labels l_i , query set $\mathcal{Q}$ instances $\bar{\mathbf{X}}_j, j \in \{1, 2, \dots, t\}$ for which the associated class label is unknown, and predicts the class label y_j for $\bar{\mathbf{X}}_j$ of each class in $\{1, 2, \dots, N\}$ to finally select the highest obtained score (Maximum A Posteriori) [26], [36], [41], i.e.,

$$\{y_1, \dots, y_t\} = f_\Theta(\mathcal{T}) = \left\{ \arg \max_{y_j \in \{1, 2, \dots, N\}} p(y_j|\mathcal{T}) \right\}_{j=1}^t \quad (2)$$

where $p(y_j|\mathcal{T})$ represents the posterior probability that the query sample $\bar{\mathbf{X}}_j$ belongs to class c given the task $\mathcal{T}$.

B. Embedding Network

Since our XAI framework is designed to be theoretically applicable to all FSL backbones (i.e., embedding networks), we take commonly used *Conv64F*, *ResNet12*, *ResNet18* and *ViT* in our experiments for shallow and deep backbones, whose architectures are shown in Section S.A. Let $\Psi_\theta(\cdot)$ represents the embedding network, θ represents the model parameters which determine the backbone. This step can be formulated as $\Psi_\theta(\mathbf{X}) = \mathbf{F}$. The image $\mathbf{X}$ is embedded to obtain the high-dimensional feature map $\mathbf{F} \in \mathbb{R}^{h \times w \times d}$, in which d represents the number of feature channels, and the spatial dimensions are $m = h \times w$, where h for height and w for width of the feature map. Here, for both *Conv64F* and *ResNet12*, when the input sizes are $84 \times 84 \times 3$, we fix the size to $h = w = 5$. Following the style of NBNN methods [32], [33], the feature map $\mathbf{F}$ is considered as $m = h \times w$ local descriptors $\{\mathbf{x}_1, \dots, \mathbf{x}_m\} \in \mathbb{R}^{1 \times 1 \times d}$ in the spatial dimensions, where the length of each local descriptor is d . Still, image-to-class measure inadvertently incorporates complex background elements and may introduce instability [33], [62], [63]. Hence, a more discriminative measure is needed.

C. Contrastive Local Descriptor

We build upon NBNN framework and introduce an image-to-multiclass measure inspired by Fisher score [64] to build a more discriminative metric through utilizing the remaining accessible samples from remaining class (See Section S.C for the theoretical connection to Fisher's LDA). For an N -way K -shot task $\mathcal{T}$, the local descriptor pool for query instance $\bar{\mathbf{X}}_j$ is defined as $\mathcal{P}_{\bar{\mathbf{X}}_j} = \bigcup_{i=1}^C \Psi_\theta(\mathbf{X}_i)$. Here, $\Psi_\theta(\mathbf{X}_i)$ represents the set of local descriptors extracted from the i th sample of the whole support set $\mathcal{S}$. For a query instance $\bar{\mathbf{X}}_j$, since the feature attribution map is made per instance, we have $t = 1$.

Hence, for this single query instance $\bar{\mathbf{X}}$, its local descriptors are denoted as $\Psi_\theta(\bar{\mathbf{X}}) = \{\mathbf{q}_1, \dots, \mathbf{q}_m\} \in \mathbb{R}^{d \times m}$. Suppose the true class of $\bar{\mathbf{X}}$ is $c \in \{1, 2, \dots, N\}$. For the following contrastive procedures, the ‘‘global’’ pool $\mathcal{P}_{\bar{\mathbf{X}}}$ here is divided into an intraclass pool $\mathcal{P}_{\bar{\mathbf{X}}}^c$ and an interclass pool $\mathcal{P}_{\bar{\mathbf{X}}}^{c'}$ (Associated with intraclass support sample(s) $\mathcal{S}^c$ and interclass support samples $\mathcal{S}^{c'}$, respectively). The contrastive similarity metric $\Phi_c(\cdot)$ for a query local descriptor $\mathbf{q}_n$ with respect to c is defined as the difference in contrast between *intraclass similarity* and *interclass similarity*, i.e.,

$$\begin{aligned} \Phi_c(\mathbf{q}_n) &= \underbrace{\frac{1}{k} \sum_{p=1}^k \cos(\mathbf{q}_n, \mathbf{x}_p^c)}_{\text{Intraclass Similarity}} - \lambda \cdot \underbrace{\frac{1}{k} \sum_{c' \neq c} \sum_{p=1}^k \cos(\mathbf{q}_n, \mathbf{x}_p^{c'})}_{\text{Interclass Similarity}} \\ z_{\text{contrast}}^{\text{ref}}(\bar{\mathbf{X}}, c) &= \sum_{n=1}^m \Phi_c(\mathbf{q}_n) \\ \cos(\mathbf{q}_n, \mathbf{x}_p^{c'}) &= \frac{\mathbf{q}_n^\top \mathbf{x}_p^{c'}}{\|\mathbf{q}_n\| \cdot \|\mathbf{x}_p^{c'}\|} \end{aligned} \quad (3)$$

where $\mathbf{x}_p^c$ is the p th k -NN local descriptor intraclass, $\mathbf{x}_p^{c'}$ is the p th k -NN from nontarget class c' (i.e., interclass), $\cos(\cdot)$ indicates the cosine similarity, and λ is a weighting factor that governs the impact of interclass similarity. When $\lambda = 0$, the framework reduces to the NBNN approach (We take $\lambda = 1$). Successively, the total contrastive similarity $z_{\text{contrast}}^{\text{ref}}(\cdot)$ for the query sample $\bar{\mathbf{X}}$ with respect to c is the sum of the contrastive similarities of all its local descriptors. Considering perturbation, for a given query instance $\bar{\mathbf{X}}$, the contrastive feature attribution score $\Delta z_{\text{contrast}}^{(a)}$ for the a th perturbed patch with respect to c is denoted as follows:

$$\Delta z_{\text{contrast}}^{(a)} = z_{\text{contrast}}^{\text{ref}}(\bar{\mathbf{X}}, c) - z_{\text{contrast}}(\bar{\mathbf{X}}'_a, c) \quad (4)$$

where $\bar{\mathbf{X}}'_a$ is the image obtained by perturbing the a th patch in $\bar{\mathbf{X}}$. Perturbed samples are generated by applying minimal modifications to individual pixels (Or local patches) across query instances [25], and the original query $\bar{\mathbf{X}}$ gives a reference score. Equation (3) provides an image-to-multiclass metric derived from Fisher's discriminant analysis.

Our C3A framework introduces three key advancements over ProtoShotXAI for few-shot XAI: 1) C3A avoids over-smooth and instable explanation by using contrastive local descriptors instead of prototypes, which makes it works for FSL backbones; 2) C3A does not need to access the weights of the class node, making it more consistent across FSL architectures (various FSL methods eschew a parameterized classification layer [26], [32], [33]); and 3) C3A is a more flexible framework where the interclass part of (3) can also be changed to any class in a episode, make it applicable for episodic explanations (See Section III-E).

D. Interpreting FSL Queries

The C3A framework can be used in many ways to explore the feature space of the FSL architectures. The C3A architecture receives two sets of inputs. One set is the support input features to compare against, and this support generates

Algorithm 1 C3A for Query Feature Attribution Maps

Given: A set of support images (divided with respect to class c to a intraclass set and a interclass set), a query image $\bar{\mathbf{X}}$ under test, average RGB pixel vector $\mathbf{R}_{RGB}$, a classification model backbone $\Psi_\theta(\cdot)$ with its parameters θ frozen.

Output: Feature attribution map $\mathbf{z}$.

- 1: $z_{\text{contrast}}^{\text{ref}}(\bar{\mathbf{X}}, c) = \sum_{n=1}^m \Phi_c(\mathbf{q}_n)$ Ref. score of $\bar{\mathbf{X}}$ with respect to c
- 2: **for** $u \leftarrow 1, 2, \dots, U$ **do** Loop through columns of the image
- 3: **for** $v \leftarrow 1, 2, \dots, V$ **do** Loop through rows of the image
- 4: $a = v + (u - 1)V$ A perturbation index for each pixel
- 5: $\bar{\mathbf{X}}'_a = \bar{\mathbf{X}}$ Init. $\bar{\mathbf{X}}'_a$ as $\bar{\mathbf{X}}$ before each perturbation
- 6: $\bar{\mathbf{X}}'_a(u, v) = \mathbf{R}_{RGB}$ Set pixel patch to avg. RGB value
- 7: $\mathbf{z}(u, v) = z_{\text{contrast}}^{\text{ref}}(\bar{\mathbf{X}}, c) - z_{\text{contrast}}(\bar{\mathbf{X}}'_a, c)$ Score for $\bar{\mathbf{X}}'$
- 8: **end for**
- 9: **end for**

an intraclass support set and an interclass set with respect to c , which are, respectively, associated with $\mathcal{P}_{\bar{\mathbf{X}}}^c$ and $\mathcal{P}_{\bar{\mathbf{X}}}^{c'}$ after embedding (for class-specific explanations, the interclass set can be changed to a specific comparison class). The other can be any set of feature alterations in a query sample under test (e.g., *adding perturbations, adding noise, and masking*). The change in contrastive similarity score, i.e., (4), provides a measure for the effects of the feature perturbation with respect to c . That is, changing features that are more important to the architecture will have a larger effect on the change to the contrastive similarity score.

E.g., feature attribution maps are generated analogously to LIME, SHAP, Grad-CAM, and ProtoShotXAI for images, as shown in Fig. 3 and formalized in Algorithm 1. The algorithm starts to select input support image(s) (e.g., images of the entire support set $\mathcal{S}$ in Fig. 3) and a test image (e.g., an image containing both the Tank and the French Bulldog in Fig. 3). Then, the support images are divided into an intraclass support set and an interclass set with respect to c . The intraclass set contains the image(s) of class c in $\mathcal{S}$, while the interclass set can be images of the rest of the classes in $\mathcal{S}$ or image(s) for another specific class for a flexible explanation. A baseline contrastive similarity metric $z_{\text{contrast}}^{\text{ref}}(\bar{\mathbf{X}}, c)$ for reference. For each pixel, the algorithm evaluates local patches (i.e., the small regions surrounding each pixel) by replacing RGB values with their average $\mathbf{R}_{RGB}$ and computing the resulting contrastive similarity score. Here, the size of the patches is set manually. Attribution significance is determined by the difference between this score and the reference. A pixel is more significant when the contrastive similarity score decreases and *vice versa*.

E. Explaining Episodic Learning

Until now, all our methods are from a testing/query perspective (explaining a model with respect to a specific class c on a test/query sample), which is applicable for both traditional ML and FSL. However, to get a more comprehensive insight into FSL, it is the same important to understand

Algorithm 2 C3A-E for Episodic Explanation Heatmaps

Given: A set of support images (N -way I -shot) (divided with respect to class c to a intraclass set and a interclass set), a query $\bar{\mathbf{X}}$ image under test, average RGB pixel vector $\mathbf{R}_{RGB}$, a classification model backbone $\Psi_\theta(\cdot)$ with its parameters θ frozen.

Output: Episodic explanation E .

- 1: $E \leftarrow \emptyset$ Init. explanation E
- 2: **for** $\mathbf{X}_i \in \mathcal{S}$ **do** For each support sample.
- 3: $z_{\text{contrast}}^{\text{ref}}(\mathbf{X}_i, c) = \sum_{n=1}^m \Phi_c(\mathbf{x}_{i,n})$ Ref. score of $\mathbf{X}_i$ with respect to c
- 4: **for** $u \leftarrow 1, 2, \dots, U$ **do** Loop through columns of the image
- 5: **for** $v \leftarrow 1, 2, \dots, V$ **do** Loop through rows of the image
- 6: $a = v + (u - 1)V$ A perturbation index for each pixel
- 7: $\mathbf{X}'_{i,a} = \mathbf{X}_i$ Init. $\mathbf{X}'_{i,a}$ as $\mathbf{X}_{i,a}$ before each perturbation
- 8: $\mathbf{X}'_{i,a}(u, v) = \mathbf{R}_{RGB}$ Set pixel patch to avg. RGB value
- 9: $\mathbf{z}_i(u, v) = z_{\text{contrast}}^{\text{ref}}(\mathbf{X}_i, c) - z_{\text{contrast}}(\mathbf{X}'_{i,a}, c)$
- 10: **end for** Accomplish calculating score for each $\mathbf{X}'_i$
- 11: $E \leftarrow E \cup \{\mathbf{z}_i\}$ Add heatmap to explanation.
- 12: **end for**
- 13: **end for**

how the model classifies novel classes given only a few support samples per class, i.e. explaining a FSL model with respect to a query instance $\bar{\mathbf{X}}$ on a support set $\mathcal{S}$. Hence, we further proposed *ProtoShot* for Episodes (ProtoShot*-E)* and *C3A for Episodes (C3A-E)* to achieve this through the previous proposed ProtoShot* and C3A. As shown in Algorithm 2, for every sample in the support set $\mathbf{X}_i$, an attribution map is calculated similar to the one in Section III-D. The key difference is that $\mathbf{X}_i$ takes the place of the previous query instance and $\bar{\mathbf{X}}$ forms $\mathcal{S}^c$, while other support samples form $\mathcal{S}^{c'}$.

F. Key Distinctions With Respect to State-of-the-Art

In Table I, we summarize key distinctions between our approaches (ProtoShot* and C3A), and state-of-the-art XAI methods. Unlike Grad-CAM, our methods provide counterfactual explanations that indicate negative correlations between inputs and classifier outputs. Concerning attribution weights on images, SHAP and Grad-CAM generate coarse heatmaps rather than pixel-level explanations (i.e., one-weight-to-one-pixel). For segmentation-based methods [LIME and SINEX(C)], the results are nondeterministic. Most approaches, including ours, are few-shot classifier-agnostic, with LIME offering broader applicability beyond DNNs. In particular, both ProtoShot* and C3A support episodic explanations, highlighting their comprehensive utility for FSL explainability. In this article, episodic explanation stands for feature-level explanations on the support samples for model output, i.e. which part of each support sample contributes to the classification results, and *vice versa*.

TABLE I

TABULAR COMPARISON OF OUR APPROACHES (MODIFIED PROTOSHOTXAI (I.E., PROTOSHOT*) AND C3A) WITH RESPECT TO STATE-OF-THE-ART METHODS EVALUATED IN THIS ARTICLE. FELL RES.: FULL RESOLUTION ATTRIBUTION MAP. DETERMIN.: DETERMINISTIC. FEW-SHOT CLF. AGNOSTIC: FEW-SHOT CLASSIFIER-AGNOSTIC. EPISODIC EXPL.: EPISODIC EXPLANATION

XAI Method /Metrics	Deep SHAP [51]	Grad-CAM [49]	LIME [50]	SIN-EX(C) [16]	ProtoShot [25]	ProtoShot* (Our Mod.)	C3A (Ours)
Counter-factuals	✓	–	✓	✓	✓	✓	✓
Pairwise	–	–	–	✓	✓	✓	✓
Full Res.	–	–	✓	–	✓	✓	✓
Determin.	✓	✓	–	–	✓	✓	✓
Few-shot Clf. Agnostic	–	✓	✓	✓	–	✓	✓
Episodic Expl.	–	✓	–	✓	–	✓	✓

IV. EXPERIMENTS

The primary goal of this section is to investigate three key questions: 1) How do traditional XAI methods, such as LIME, SHAP, and GradCAM, perform in FSL tasks? 2) What are the effects of using only image-level prototypes (i.e. ProtoShotXAI) for few-shot explainability? 3) How does our proposed C3A framework, which generates image-to-multiclass few-shot feature attribution scores, perform in FSL? To answer these questions, we analyze explanations from both the query sample and the episodic perspectives. While the current section focuses on standard FSL backbones (Conv64F and ResNet12), we extensively validate that our conclusions hold for larger architectures (ResNet18 and ViT) in the Supplementary Material, demonstrating the broad applicability of C3A.

A. Experimental Setup

1) *Datasets*: We conducted all the experiments on four benchmark datasets as follows.

miniImageNet: This dataset [42], a smaller version of ImageNet [65], comprises 100 classes, each containing 600 images, with a resolution of 84×84 pixels per image. Keeping with the splits established in [38], we allocate 64, 16, and 20 classes for training, validation, and testing, respectively.

Stanford Dogs: This dataset [66] was initially designed for fine-grained image classification and includes 120 different dog breeds, totaling 20 580 images. Here, our backbones were trained to perform a fine-grained few-shot classification task on this dataset, selecting 70 classes for training, 20 for validation, and 30 for testing.

Stanford Cars: This dataset [67] also serves as a benchmark dataset for fine-grained classification tasks and includes of 196 classes of cars, comprising a total of 16 185 images. In a similar manner, 17 for validation and 49 for testing.

CUB-200: This dataset [68] comprises 6033 images representing 200 different bird species. Similarly, we choose 130, 20, and 50 classes for training, validation, and test.

TABLE II

FEW-SHOT CLASSIFICATION RESULTS THE MINI IMAGE NET, STANFORD DOGS, STANFORD CARS, AND CUB-200 DATASETS. PROTONET[‡] WITH RESNET12 BACKBONE ACHIEVES CONSISTENT IMPROVEMENTS. “/” DENOTES NO REPORTED SCORES TO OUR KNOWLEDGE; THERE IS NO REPORTED (REP) SCORE. ALL EXPERIMENTS UNDER THE 5-WAY 5-SHOT SETTING WITH DATA AUGMENTATION

Embed.	<i>miniImageNet</i>		Stanford Dogs		Stanford Cars		CUB-200	
	REP	Ours	REP	Ours	REP	Ours	REP	Ours
Conv64F	68.56	66.37	48.19	50.65	52.93	52.80	70.77	68.17
ResNet12	75.16	67.35	/	52.85	/	36.81	/	72.61

For the last three fine-grained datasets, all images in these datasets are resized to 84×84 in line with the general-purpose dataset *miniImageNet*. To the best of our knowledge, the *miniImageNet*, Stanford Dogs, Stanford Cars, and CUB-200 datasets have not been utilized in any other XAI approaches, and the application of these four FSL benchmarks represents a novel contribution to the field.

2) *Baselines*: Different from existing studies [16], [25], we propose a comprehensive approach that considers both query explanation and episodic explanation (i.e., explanation heat maps of support samples). For evaluation, we compare C3A with multiple state-of-the-art XAI methods. Specifically, for query explanation, C3A is benchmarked against LIME,¹ GradSHAP,² Grad-CAM,³ and ProtoShot*; while for episodic explanations, C3A-E is compared with Grad-CAM, SINEX, SINEXC,⁴ and ProtoShot*.

The rational of this design include three points: 1) original ProtoShotXAI needs to access the classifier weights of DNN classifier, while most FSL model do not possess a parameterized classification layer [26], [32]; 2) SINEX(C) can only generate episodic explanations; and 3) SHAP needs access to a background set to learn Shapely values, which is not accessible in our episodic explanation setting [7]. Moreover, for episodic explanations, we did not include LIME because SINEX(C) already incorporates and extends LIME’s perturbation-based methodology specifically for FSL contexts. While LIME is effective for individual instance explanations (Hence its inclusion in query explanations), it lacks the architecture to naturally handle episodic relationships between support and query samples. SINEX(C) extends the perturbation concept to operate across the support set, making it inherently more suitable for episodic explanations.

3) *Training Settings*: All methods are trained from scratch using the Adam optimizer with an initial learning rate of 10^{-3} . Models with Conv64F and ViT backbone were trained for 100 epochs with 200 training episodes per epoch, while ResNet12 and ResNet18 models are trained for 20 epochs with 2000 training episodes per epoch (See the results of ViT and ResNet18 in Section S.D). All reported results represent top-1 classification accuracy across 1000 test episodes.

¹<https://github.com/marcotcr/lime>

²<https://github.com/shap/shap>

³<https://github.com/jacobgil/pytorch-grad-cam>

⁴<https://github.com/andreafede/SINEX>

TABLE III

QUANTITATIVE RESULTS FOR EPISODIC EXPLANATIONS OF GRAD-CAM, SINEX, SINEXC, OUR MODIFIED PROTOSHOT*-E, AND OUR C3A-E ON STANFORD DOG, STANFORD CARS, AND CUB-200 DATASETS. THE BEST ARE IN BOLD AND THE SECOND BEST VALUES ARE UNDERLINED

Embed.	<i>miniImageNet</i>		Stanford Dogs		Stanford Cars		CUB-200	
	REP	Ours	REP	Ours	REP	Ours	REP	Ours
Conv64F	68.56	66.37	48.19	50.65	52.93	52.80	70.77	68.17
ResNet12	75.16	67.35	/	52.85	/	36.81	/	72.61

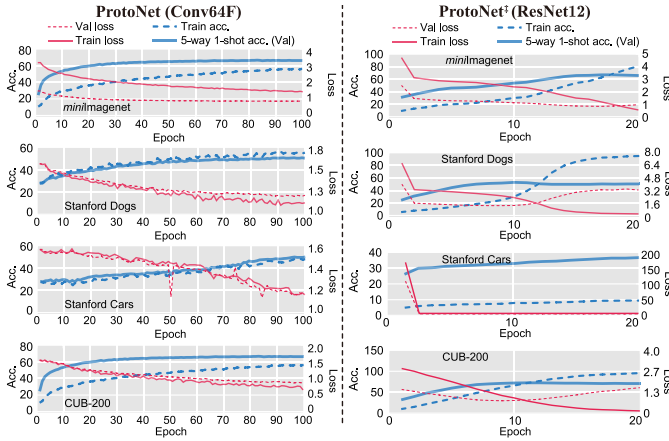


Fig. 4. Training and validation accuracy/loss curves comparison between ProtoNet and ProtoNet*. Blue curves represent accuracy metrics while red curves indicate loss values, with solid lines denoting training set performance and dashed lines corresponding to validation set results.

For data preprocessing, we implemented standard augmentation techniques, including random cropping, left-right flipping, and color jittering during both training and meta-training phases. Due to computational constraints, all backbones were trained using a 5-way 5-shot episodic training regime during meta-training, which may yield slightly lower accuracy than originally reported scores (See Table II). This difference is expected as increasing the number is a common performance enhancement technique in FSL literature [8], [26], [27].

All training-related experiments were performed using two NVIDIA RTX A4000 GPUs (16-GB memory each) with TensorFlow 2.5.0. The codes for training were based on a public repository of ProtoNet.⁵ Training convergence curves are presented in Fig. 4. It should be noted that for ProtoNet* on the Stanford Cars dataset, we encountered persistent exploding gradient issues across all feasible hyperparameter configurations within our memory constraints, with the best achievable accuracy limited to 36.81% as documented in Table III.

4) *XAI Experimental Protocol*: Our experiment protocol is to first train ProtoNet and ProtoNet* on each dataset to establish sufficient classification performance. Post-training, we integrated the C3A framework (detailed in Section III and visualized in Fig. 3). For fair comparison, all explainability methods interpreted identical images using the same model states. Visualization was standardized using SHAP's attribution methodology [51], where feature importance weights z

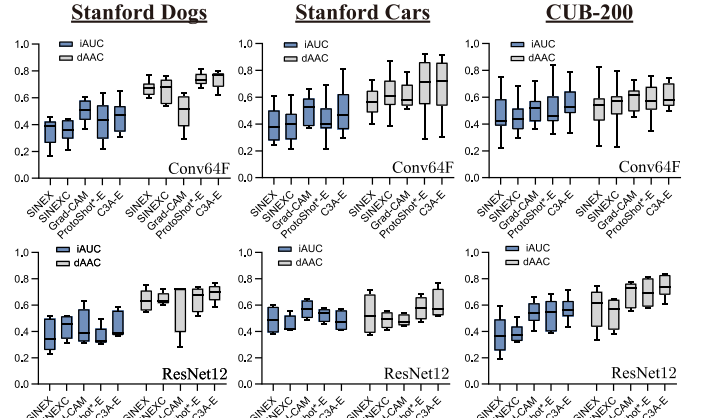


Fig. 5. Box plots of the calculated *iAUC*, *dAUC* of the comparative experiments with Conv64F and ResNet12 on Stanford Dogs, Stanford Cars, and CUB-200. The error bar shows the *max* and *min* values of each group, and the line across the box shows the median of the group. It can be observed from a distribution view that our modified ProtoShot*-E and our C3A-E generally outperform other methods.

were calculated with the 99.9% percentile of $|z|$ defining the color axis range. Thereby, positive and negative attributions were uniformly rendered in red and blue, respectively, with the percentile value setting the dynamic bounds of the color scale.

- 1) *For Query Explanations*: As shown in Algorithm 1, C3A generates attribution maps by measuring how the perturbation of individual pixels affects the contrastive score. For each location, we replace the pixel patch with the average RGB value and calculate the difference between the reference score and the perturbed score, effectively quantifying each pixel's contribution to the model's decision.
- 2) *For Episodic Explanations*: As shown in Algorithm 2, C3A-E extends this perturbation approach to all support samples in an episode. For each support image, we compute individual attribution maps following the same pixel-wise patch perturbation methodology, then aggregate these maps into a comprehensive episodic explanation that reveals feature importance across the support set.

B. Feature Attribution Experiments

Previous research by Samuel and Gregory [25] demonstrated that ProtoShotXAI can be used for few-shot character classification on the Omniglot dataset [48], while SN explanation through SINEX(C) was applied to Audio MINST dataset [16], [69]. However, real-world CV applications involve complex features or pixels beyond binary pixels or spectrograms. Our experiments qualitatively evaluate C3A on datasets with complex image features. As shown on the left of Fig. 2(a), in our first example [See Fig. 2(a)], we created an artificial input with minimal contextual cues in line with [25] (i.e. Tank and French Bulldog) for the backbone trained on *miniImageNet*. The background of this image is an average pixel value that appears as a brownish color and provides neu-

⁵<https://github.com/ulises-jeremias/prototypical-networks-tf>

tral information. Subsequently, we then tested three examples with real backgrounds using backbones trained on fine-grained FSL benchmarks, Stanford Dogs, Stanford Cars, and CUB-200, respectively. For each image, we show the attributions produced from Conv64F and ResNet12. For Grad-SHAP, Grad-CAM, and C3A, the $h = w = 5$ output feature map from preflatten layers (layer 8 for Conv64F and layer 24 for ResNet12) was taken. And for ProtoShot*, the flattened layer output of Conv64F and ResNet12 were taken. The ProtoNet output is calculated through Euclidean distances between supporting prototypes and the query for methods that consider the whole model output (e.g., LIME).

For each trained backbone and each example image in Fig. 2(a), C3A consistently highlights positive features (red) on the class of interest and negative features (blue) on the counter class. For instance, C3A showed consistent positive features for the Tank attributions around turret/armour and negative features in the region of the French Bulldog. For the French Bulldog attributions, C3A has a positive response around the face/body and a negative response around the tank. Though all methods show some inconsistencies, with respect to attribution and class (e.g., some positive feature attributions on the Tank barrel/treads for the French Bulldog class), demonstrates superior spatial consistency. This capability is not limited to shallow networks; as shown in Supplementary Figs. S3 and S4, C3A maintains superior spatial specificity and semantic alignment even on ResNet18 and ViT backbones, effectively handling their different inductive biases. The LIME output is the closest to C3A yet, with poor stability due to its inconsistent segmentation of large regions. Grad-CAM generates blurred attributions limited to positive features, while Grad-SHAP often incorrectly highlights background elements. ProtoShot* exhibits over-smoothing we mentioned previously in Section I, especially in 5-shot conditions. In conclusion, 1) traditional XAI methods show inconsistencies in FSL tasks, 2) ProtoShot* exhibits over-smoothing, while 3) C3A consistently highlights class-relevant features with superior spatial consistency.

C. Episodic Explanation Experiments

Fig. 2(b) shows the comparative results on *miniImageNet*, Stanford Dogs, Stanford Cars, and CUB-200 under 1-shot and 5-shot conditions with ResNet12. It can be observed that traditional methods still present unstable results. Notably, Grad-CAM even failed on Stanford Dog attributions. For ProtoShot*, we still observe over-smoothing especially in 5-shot conditions. Compared with those, the proposed C3A, although still has inconsistency, outperforms other comparative methods with better specificity with respect to classes.

Moreover, from a distribution perspective, we also provide a box plot for quantitative results in our episodic experiments. As shown in Fig. 5, our modified ProtoShot*-E and proposed C3A-E exhibit significant improvement versus other comparative methods, which is consistent with results in Table III. The concentrated distribution of sample points around central statistic values further confirms the consistency and robustness of our results across different episodes, strengthening the validity of our approach.

TABLE IV
QUANTITATIVE ABLATION STUDY ON STANFORD DOG, STANFORD CARS, AND CUB-200 DATASETS OF C3A-E VERSUS ITS ABLATION VARIANTS. THE BEST RESULTS ARE IN BOLD AND THE CLOSEST ONES ARE UNDERLINED

XAI Method	FSL Embed.	Stanford Dogs		Stanford Cars		CUB-200	
		<i>iAUC</i>	<i>dAAC</i>	<i>iAUC</i>	<i>dAAC</i>	<i>iAUC</i>	<i>dAAC</i>
w/o Local	Conv64F	0.42	<u>0.74</u>	0.43	0.65	0.51	0.58
w/o Contra.	Conv64F	± 0.15	± 0.04	± 0.12	± 0.21	± 0.14	± 0.12
w/o Local	Conv64F	0.43	<u>0.74</u>	<u>0.47</u>	0.67	0.55	0.63
w/ Contra.	Conv64F	± 0.14	± 0.06	± 0.13	0.21	± 0.09	0.12
w/ Local	Conv64F	0.48	0.76	0.45	0.67	0.57	<u>0.62</u>
w/o Contra.	Conv64F	± 0.13	± 0.05	± 0.12	± 0.17	± 0.11	± 0.10
C3A-E (Ours)	Conv64F	<u>0.46</u>	<u>0.74</u>	0.49	0.66	<u>0.56</u>	0.61
		± 0.12	± 0.06	± 0.15	± 0.21	± 0.12	± 0.09
w/o Local	ResNet12	0.36	0.64	<u>0.53</u>	0.58	0.53	0.69
w/o Contra.	ResNet12	± 0.08	± 0.09	± 0.05	± 0.09	± 0.12	± 0.10
w/o Local	ResNet12	0.36	0.65	0.56	0.57	0.50	<u>0.71</u>
w/ Contra.	ResNet12	± 0.07	± 0.08	± 0.03	± 0.07	± 0.11	± 0.11
w/ Local	ResNet12	0.36	<u>0.63</u>	0.51	0.61	0.58	0.70
w/o Contra.	ResNet12	± 0.07	± 0.12	± 0.06	± 0.11	± 0.11	± 0.14
C3A-E (Ours)	ResNet12	0.45	0.69	0.48	<u>0.60</u>	<u>0.57</u>	0.74
		± 0.10	± 0.07	± 0.08	± 0.11	± 0.09	± 0.08

Fig. 6 shows an example task of explanation returned by C3A-E versus comparative methods on Stanford Cars. Similar to the last session, light blue indicates a negative influence and pink shows a positive influence on similarity scores. While SINEX(C) suffers from inconsistent segmentation, for the gradient-based method, Grad-CAM fails on low similarity classes. The ProtoShot*-E and C3A-E produce the most consistent attribution maps. Following [16], [52], [70], we quantify explanation quality using insertion/deletion metrics, where higher *insertion-area-under-curve* (*iAUC*) and *deletion-area-above-curve* (*dAAC*) values indicate better explanations that successfully identify pixels critical for classification [52]. As shown in the right panel of Fig. 6, C3A-E consistently outperforms competing methods. Notably, Grad-CAM exhibits a “reversed” deletion curve, revealing its misleading attributions in few-shot episodes. ProtoShot*-E emerges as the closest competitor, while SINEX(C)’s performance appears limited by over-reliance on segmentation. These quantitative results confirm our method’s superior ability to identify discriminative features in FSL contexts. Moreover, quantitative evaluation across 600 experiments on Stanford Dogs, Stanford Cars, and CUB-200 with 60 randomly sampled tasks in total (20 for each dataset) confirms C3A-E’s superiority. To ensure robustness, we extended this evaluation to ResNet18 and ViT backbones (See quantitative results in Supplementary Table S6). For each method, we run 80 experiments (20 for Conv64F, ResNet12, ResNet18 and ViT, respectively), measuring the overall *iAUC* and *dAAC* mean and S.D. values. Table III shows the resulting *iAUC* and *dAAC* values for Conv64F and ResNet12, while the Supplementary results confirm consistent performance gains on the larger backbones. For a fair comparison, we make $l_1 = y = 0$ to maintain Grad-CAM functionality. Our C3A-E achieves improved *iAUC* and *dAAC* at the same time with maximum absolute gains of 19.72% and 25.88%, respectively, over existing methods. For our three key questions, our episodic evaluation shows: 1) traditional XAI methods struggle with

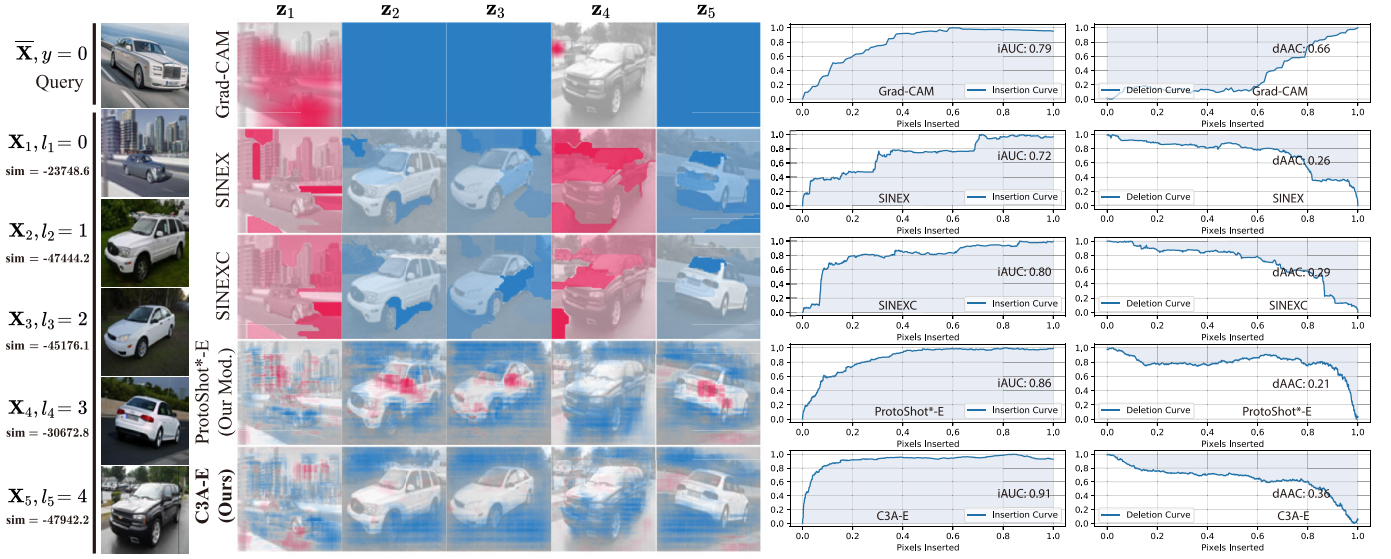


Fig. 6. Comparison of episodic explanation produced by Grad-CAM, SINEX, SINEXC, ProtoShot*-E (Our Mod.), and C3A-E (Ours) of a correct 5-way 1-shot classification on Rolls-Royce Phantom Sedan 2012 ($y = l_1 = 0$). The rest four categories ($l_2, \dots, l_4 = 1, \dots, 4$) are Buick Rainier SUV 2007, Ford Focus Sedan 2007, Chevrolet TrailBlazer SS 2009 and Audi S4 Sedan 2012. From the attribution maps on the left, it can be observed that our C3A-E and ProtoShot*-E showed more fine-grained attributions. Also, the $iAUC$ and $dAAC$ calculated from the curves on the right also show our methods' fidelity to the model's classification results.

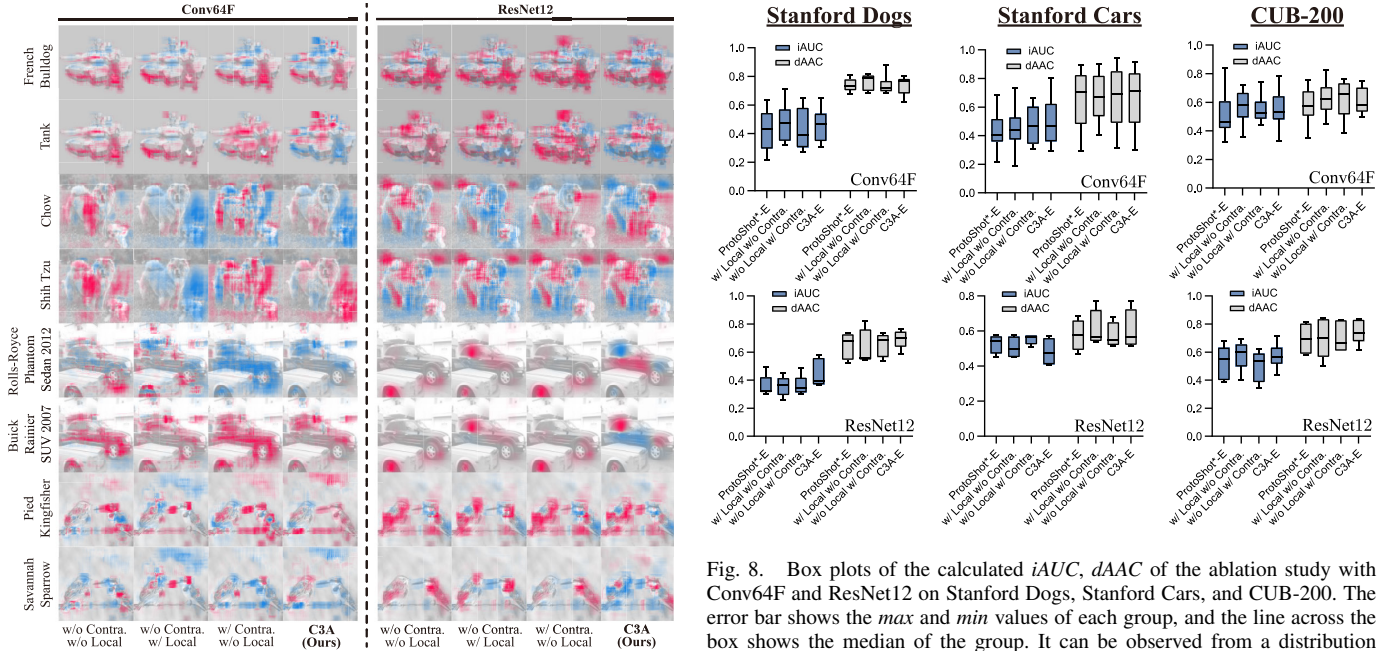


Fig. 7. Qualitative results of the ablation study. The query images and Conv64F, ResNet12 backbones are the same as Fig. 2(a) and (b). The result w/o both contra. w/o local is from ProtoShot*. Ours results w/o contra. tend to show identical explanations with different classes, and results w/o local tend to lack sensitivity on different regions. Meanwhile, our C3A overcome these issues, presenting more consistent attributions.

FSL tasks; 2) ProtoShot*-E shows limitations; while 3) C3A-E achieves significant improvements in both $iAUC$ and $dAAC$ across diverse datasets.

D. Model Sensitivity Evaluation

1) *Ablation Study*: To verify the effectiveness of our image-to-multiclass approach, we conducted an ablation study with

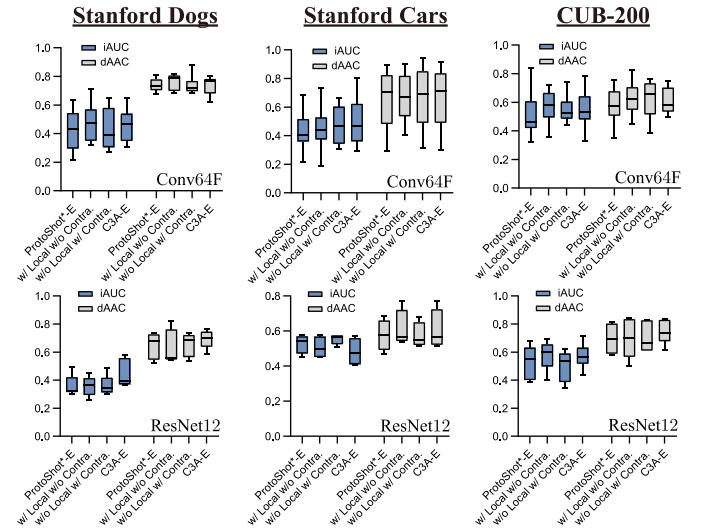


Fig. 8. Box plots of the calculated $iAUC$, $dAAC$ of the ablation study with Conv64F and ResNet12 on Stanford Dogs, Stanford Cars, and CUB-200. The error bar shows the max and min values of each group, and the line across the box shows the median of the group. It can be observed from a distribution perspective that the ablation variants produce competitive results, and all of our improvements outperform ProtoShot*-E.

four variants: 1) C3A(-E) (Proposed); 2) w/o Local w/ Contra. (Using global prototypes with contrastive metrics); and 3) w/ Local w/o Contra. (Using only local descriptors with k -NN), and w/o Local w/o Contra. [i.e., ProtoShot*(-E)]. Results across all four benchmarks under 5-shot settings consistently demonstrate that our combined approach outperforms both single-component variants. ProtoShot*(-E), which uses neither local nor contrastive components, performs the worst overall. These findings confirm that the integration of both local descriptors and contrastive measure-

TABLE V

QUANTITATIVE SENSITIVITY STUDY FOR C3A-E ACROSS THREE KEY HYPERPARAMETERS: (A) PERTURBATION PATCH SIZE, (B) NUMBER OF k -NN NEIGHBORS, AND (C) NUMBER OF SUPPORT SHOTS. THE ANALYSIS IS PERFORMED ON CONV64F AND RESNET12 BACKBONES ACROSS THREE DATASETS. FOR EACH EXPERIMENT, THE BEST RESULT IS SHOWN IN BOLD AND THE SECOND BEST IS UNDERLINED

Parameter Value	FSL Embed.	Stanford Dogs		Stanford Cars		CUB-200	
		<i>iAUC</i>	<i>dAAC</i>	<i>iAUC</i>	<i>dAAC</i>	<i>iAUC</i>	<i>dAAC</i>
(a) Sensitivity to Perturbation Patch Size							
1	Conv64F	0.41 ± 0.15	0.67 ± 0.08	0.42 ± 0.14	0.59 ± 0.19	0.49 ± 0.16	0.55 ± 0.12
2	Conv64F	0.44 ± 0.12	0.70 ± 0.07	0.45 ± 0.13	0.62 ± 0.20	0.51 ± 0.14	0.57 ± 0.11
3	Conv64F	0.45 ± 0.11	0.72 ± 0.06	0.47 ± 0.13	0.64 ± 0.19	0.54 ± 0.13	0.59 ± 0.10
4	Conv64F	0.46 ± 0.12	0.74 ± 0.06	0.49 ± 0.15	0.66 ± 0.21	0.56 ± 0.12	0.61 ± 0.09
5	Conv64F	0.44 ± 0.13	0.71 ± 0.06	0.46 ± 0.14	0.63 ± 0.18	0.53 ± 0.13	0.58 ± 0.11
6	Conv64F	0.42 ± 0.14	0.69 ± 0.07	0.45 ± 0.15	0.62 ± 0.22	0.52 ± 0.14	0.57 ± 0.12
7	Conv64F	0.40 ± 0.15	0.67 ± 0.08	0.43 ± 0.16	0.60 ± 0.23	0.50 ± 0.15	0.55 ± 0.13
8	Conv64F	0.39 ± 0.16	0.66 ± 0.09	0.42 ± 0.17	0.59 ± 0.24	0.49 ± 0.16	0.54 ± 0.14
9	Conv64F	0.38 ± 0.17	0.65 ± 0.10	0.41 ± 0.18	0.58 ± 0.25	0.48 ± 0.17	0.53 ± 0.15
10	Conv64F	0.37 ± 0.18	0.64 ± 0.11	0.40 ± 0.19	0.57 ± 0.26	0.47 ± 0.18	0.52 ± 0.16
1	ResNet12	0.38 ± 0.13	0.62 ± 0.10	0.41 ± 0.11	0.54 ± 0.14	0.50 ± 0.14	0.67 ± 0.11
2	ResNet12	0.41 ± 0.12	0.65 ± 0.09	0.44 ± 0.10	0.56 ± 0.13	0.53 ± 0.13	0.70 ± 0.10
3	ResNet12	0.43 ± 0.11	0.67 ± 0.08	0.46 ± 0.09	0.60 ± 0.11	0.55 ± 0.12	0.72 ± 0.09
4	ResNet12	0.45 ± 0.10	0.69 ± 0.07	0.48 ± 0.08	0.59 ± 0.12	0.57 ± 0.09	0.74 ± 0.08
5	ResNet12	0.44 ± 0.11	0.68 ± 0.08	0.47 ± 0.09	0.58 ± 0.12	0.56 ± 0.10	0.73 ± 0.09
6	ResNet12	0.42 ± 0.12	0.66 ± 0.09	0.46 ± 0.09	0.57 ± 0.13	0.54 ± 0.12	0.71 ± 0.10
7	ResNet12	0.40 ± 0.13	0.64 ± 0.10	0.43 ± 0.11	0.55 ± 0.14	0.52 ± 0.13	0.69 ± 0.11
8	ResNet12	0.39 ± 0.14	0.63 ± 0.11	0.42 ± 0.12	0.54 ± 0.15	0.51 ± 0.14	0.68 ± 0.12
9	ResNet12	0.37 ± 0.15	0.61 ± 0.12	0.41 ± 0.13	0.53 ± 0.16	0.49 ± 0.15	0.66 ± 0.13
10	ResNet12	0.36 ± 0.16	0.60 ± 0.13	0.39 ± 0.14	0.52 ± 0.17	0.48 ± 0.16	0.65 ± 0.14
(b) Sensitivity to Number of k -NN Neighbors							
1	Conv64F	0.32 ± 0.19	0.59 ± 0.14	0.36 ± 0.21	0.52 ± 0.29	0.42 ± 0.21	0.47 ± 0.18
2	Conv64F	0.34 ± 0.18	0.61 ± 0.13	0.38 ± 0.20	0.55 ± 0.28	0.44 ± 0.20	0.50 ± 0.17
3	Conv64F	0.36 ± 0.18	0.64 ± 0.12	0.40 ± 0.19	0.57 ± 0.27	0.46 ± 0.19	0.52 ± 0.16
5	Conv64F	0.39 ± 0.16	0.67 ± 0.10	0.43 ± 0.17	0.59 ± 0.25	0.49 ± 0.17	0.54 ± 0.14
7	Conv64F	0.42 ± 0.15	0.75 ± 0.08	0.45 ± 0.16	0.64 ± 0.24	0.54 ± 0.14	0.57 ± 0.13
9	Conv64F	0.44 ± 0.14	0.72 ± 0.07	0.47 ± 0.15	0.62 ± 0.23	0.52 ± 0.14	0.59 ± 0.12
10	Conv64F	0.46 ± 0.12	0.74 ± 0.06	0.49 ± 0.15	0.66 ± 0.21	0.56 ± 0.12	0.61 ± 0.09
11	Conv64F	0.43 ± 0.14	0.71 ± 0.08	0.46 ± 0.17	0.63 ± 0.23	0.53 ± 0.14	0.58 ± 0.12
13	Conv64F	0.41 ± 0.15	0.69 ± 0.09	0.44 ± 0.18	0.61 ± 0.24	0.51 ± 0.15	0.56 ± 0.13
15	Conv64F	0.40 ± 0.16	0.68 ± 0.10	0.43 ± 0.19	0.60 ± 0.25	0.50 ± 0.16	0.55 ± 0.14
1	ResNet12	0.31 ± 0.18	0.55 ± 0.15	0.34 ± 0.17	0.46 ± 0.20	0.43 ± 0.18	0.60 ± 0.16
2	ResNet12	0.33 ± 0.17	0.57 ± 0.14	0.36 ± 0.16	0.49 ± 0.19	0.45 ± 0.17	0.62 ± 0.15
3	ResNet12	0.36 ± 0.16	0.60 ± 0.13	0.38 ± 0.15	0.51 ± 0.18	0.47 ± 0.16	0.64 ± 0.14
5	ResNet12	0.38 ± 0.14	0.62 ± 0.11	0.41 ± 0.13	0.53 ± 0.16	0.58 ± 0.13	0.67 ± 0.12
7	ResNet12	0.40 ± 0.13	0.65 ± 0.10	0.49 ± 0.11	0.58 ± 0.14	0.52 ± 0.13	0.70 ± 0.11
9	ResNet12	0.42 ± 0.12	0.66 ± 0.09	0.45 ± 0.11	0.56 ± 0.13	0.54 ± 0.12	0.71 ± 0.10
10	ResNet12	0.45 ± 0.10	0.69 ± 0.07	0.48 ± 0.08	0.60 ± 0.11	0.57 ± 0.09	0.74 ± 0.08
11	ResNet12	0.42 ± 0.12	0.66 ± 0.09	0.45 ± 0.11	0.57 ± 0.13	0.54 ± 0.12	0.71 ± 0.10
13	ResNet12	0.41 ± 0.13	0.65 ± 0.10	0.44 ± 0.12	0.56 ± 0.14	0.53 ± 0.13	0.70 ± 0.11
15	ResNet12	0.39 ± 0.14	0.64 ± 0.11	0.42 ± 0.13	0.55 ± 0.15	0.51 ± 0.14	0.68 ± 0.12
(c) Sensitivity to Number of Support Shots							
1	Conv64F	0.45 ± 0.13	0.73 ± 0.07	0.48 ± 0.16	0.65 ± 0.22	0.55 ± 0.13	0.60 ± 0.10
2	Conv64F	0.43 ± 0.13	0.71 ± 0.07	0.46 ± 0.17	0.63 ± 0.22	0.53 ± 0.13	0.58 ± 0.11
3	Conv64F	0.43 ± 0.14	0.71 ± 0.08	0.45 ± 0.18	0.62 ± 0.23	0.52 ± 0.14	0.58 ± 0.12
4	Conv64F	0.44 ± 0.13	0.72 ± 0.07	0.47 ± 0.17	0.64 ± 0.22	0.54 ± 0.13	0.59 ± 0.11
5	Conv64F	0.46 ± 0.12	0.74 ± 0.06	0.49 ± 0.15	0.66 ± 0.21	0.56 ± 0.12	0.61 ± 0.09
6	Conv64F	0.42 ± 0.14	0.70 ± 0.08	0.45 ± 0.18	0.62 ± 0.23	0.52 ± 0.14	0.57 ± 0.12
7	Conv64F	0.42 ± 0.15	0.70 ± 0.09	0.44 ± 0.19	0.61 ± 0.24	0.51 ± 0.15	0.57 ± 0.13
8	Conv64F	0.41 ± 0.15	0.69 ± 0.09	0.44 ± 0.19	0.61 ± 0.24	0.51 ± 0.15	0.56 ± 0.13
9	Conv64F	0.41 ± 0.16	0.68 ± 0.10	0.43 ± 0.20	0.60 ± 0.25	0.50 ± 0.16	0.55 ± 0.14
10	Conv64F	0.40 ± 0.16	0.68 ± 0.10	0.43 ± 0.20	0.60 ± 0.25	0.50 ± 0.16	0.55 ± 0.14
1	ResNet12	0.44 ± 0.11	0.68 ± 0.08	0.47 ± 0.09	0.59 ± 0.12	0.56 ± 0.10	0.73 ± 0.09
2	ResNet12	0.42 ± 0.11	0.66 ± 0.08	0.45 ± 0.10	0.57 ± 0.12	0.54 ± 0.11	0.71 ± 0.09
3	ResNet12	0.42 ± 0.12	0.66 ± 0.09	0.45 ± 0.11	0.57 ± 0.13	0.54 ± 0.12	0.71 ± 0.10
4	ResNet12	0.43 ± 0.11	0.67 ± 0.08	0.46 ± 0.10	0.58 ± 0.12	0.55 ± 0.11	0.72 ± 0.09
5	ResNet12	0.45 ± 0.10	0.69 ± 0.07	0.48 ± 0.08	0.60 ± 0.11	0.57 ± 0.09	0.74 ± 0.08
6	ResNet12	0.41 ± 0.12	0.65 ± 0.09	0.44 ± 0.11	0.56 ± 0.13	0.53 ± 0.12	0.70 ± 0.10
7	ResNet12	0.41 ± 0.13	0.65 ± 0.10	0.44 ± 0.12	0.56 ± 0.14	0.53 ± 0.13	0.70 ± 0.11
8	ResNet12	0.40 ± 0.13	0.64 ± 0.10	0.43 ± 0.12	0.55 ± 0.14	0.52 ± 0.13	0.69 ± 0.11
9	ResNet12	0.40 ± 0.14	0.64 ± 0.11	0.43 ± 0.13	0.55 ± 0.15	0.51 ± 0.14	0.68 ± 0.12
10	ResNet12	0.39 ± 0.14	0.63 ± 0.11	0.42 ± 0.13	0.54 ± 0.15	0.51 ± 0.14	0.68 ± 0.12

ments is essential for effective FSL explanations. Qualitative results of the ablation study are visually demonstrated

in Fig. 8. Following the protocol established in Section III-E, we conducted 240 additional experiments across

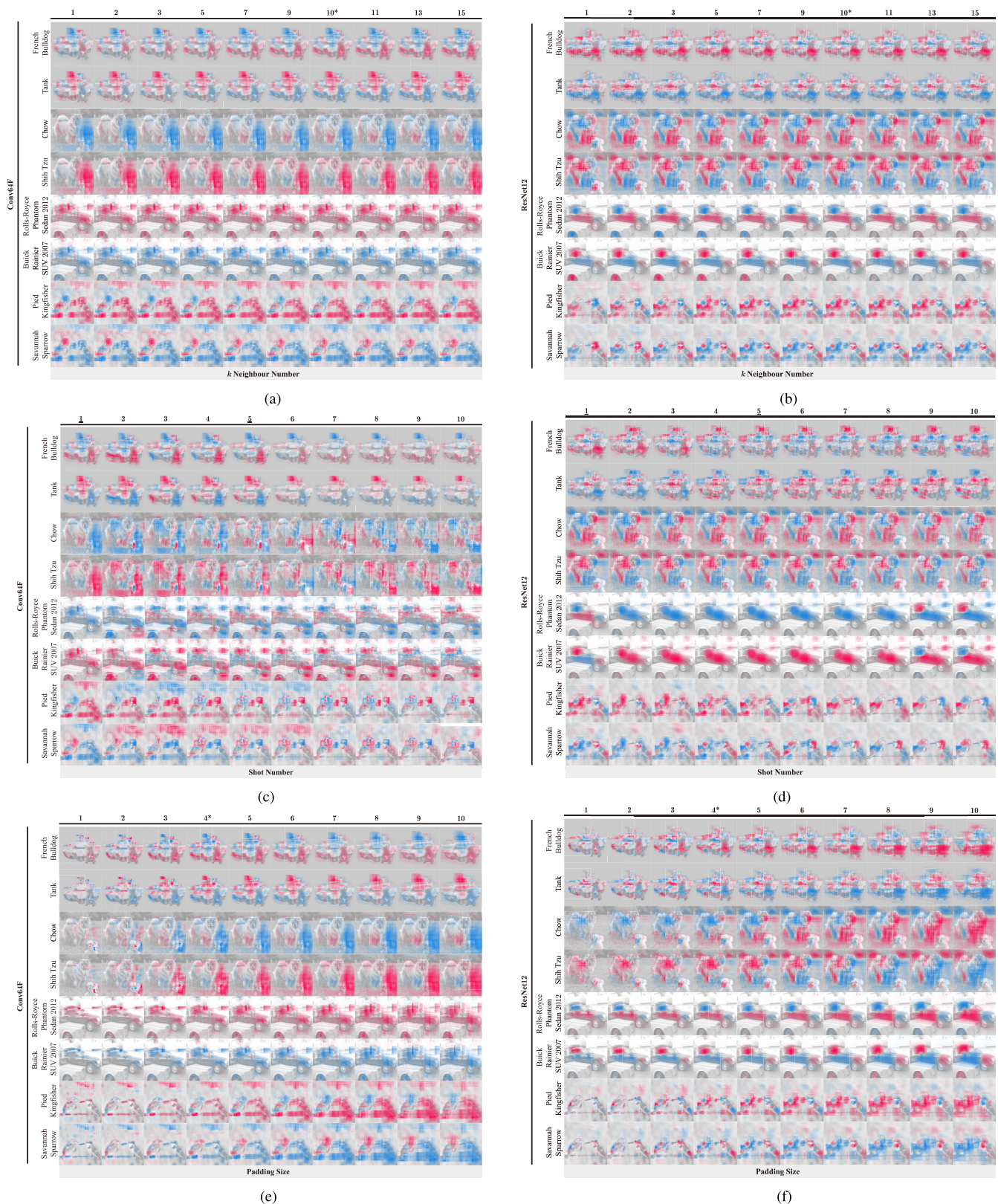


Fig. 9. C3A feature attributions on Conv64F and ResNet12 across various datasets. (a) and (b) Compare attributions with respect to the number of neighbors, showing increased consistency with more neighbors. (c) and (d) Effect of varying the number of support shots, with optimal performance generally between 1 and 5-shot. (e) and (f) Tradeoff with different perturbation area sizes, balancing attribution coverage and detail.

ablation variants on Stanford Dogs, Stanford Cars, and CUB-200 datasets. These experiments span 60 randomly sampled tasks (20 per dataset) using both Conv64F and ResNet12 backbones (20 experiments each). Table IV summarizes the

mean and S.D for each variant compared to our proposed C3A. The results clearly demonstrate that integrating either NBN or Fisher's discriminant significantly improves performance over the baseline ProtoShot*-E configuration (w/o Local w/o

Contra.). The statistical distribution analysis in Fig. 8 further confirms these findings, confirming the effectiveness of our architectural design choices.

2) *Hyperparameter Study*: C3A(-E) involves three key hyperparameters for attribution map generation: 1) the number of neighbors k ; 2) the size of perturbed local patches; and 3) the number of support samples per class. We perform a systematic analysis by independently varying each parameter to evaluate its impact across diverse images. Based on qualitative assessment of Fig. 9, we identify an optimal configuration ($k = 10$, patch padding of 6, and 1 support samples per class), which is consistent with ProtoShot* and ablation variants throughout our experiments for fair comparison. Comprehensive quantitative results supporting these observations are presented in Table V, demonstrating the robustness of C3A(-E) around these chosen values across multiple datasets and backbones. Crucially, this hyperparameter stability is not unique to shallower networks; as detailed in Supplementary Table S7 and Fig. S5, the same optimal configurations generalize effectively to ResNet18 and ViT, indicating that C3A does not require extensive per-architecture tuning. If not noted specifically, we generally follow this set.

Number of Neighbors: The k -NN process in (3) introduces the number of neighbors k as a hyperparameter for C3A similarity score calculation. With embedding dimensions of 5×5 yielding 25 local descriptors in our experiments, we select $k = 10$ as an optimal tradeoff between measurement stability and computational efficiency. As illustrated in Fig. 9, performance consistency generally improves slightly as k increases from 1 to 15. This trend is quantitatively supported by the iAUC and dAAC metrics presented in Table V, where $k = 10$ consistently yields strong or best performance across various settings, validating our choice. Supplementary results confirm that this stabilization effect of increasing k is universal across all evaluated backbones.

Padding Size: The local patch is defined as a square region centered around a pixel of interest, with the entire area replaced by the reference pixel's average RGB value to induce feature manifold deviations. We implemented this as a moving window across the image, evaluating each pixel as a center of interest. The padding size determines the spatial extent of the patch in both vertical and horizontal dimensions. Fig. 9 demonstrates the effect of increasing the padding size from 1 to 10 for Conv64F and ResNet12 across four benchmarks. Larger padding sizes produce more comprehensive object coverage but reduce fine-grained details. For networks operating on 84×84 RGB images, we empirically determined that a padding size of 6 provides the optimal balance between coverage and detail preservation. The quantitative analysis in Table V corroborates this, showing that a padding of 6 achieves a favorable balance in iAUC/dAAC scores compared to smaller or excessively larger padding sizes. Notably, this tradeoff between coverage and detail remains consistent even for the ViT [See Supplementary Fig. S5(e) and (f)], despite its distinct patch-based processing mechanism.

Number of Support Samples: C3A(-E) facilitates image-to-multiclass comparisons by representing query images as

combinations of local descriptors generated from class support sample embeddings. As shown in Fig. 9, C3A performance initially improves with additional shots but degrades beyond 5-shot. This aligns with the findings of existing NBNN-based FSL literature [33], as excessive local descriptors can be contaminated by background information. Notably, this limitation is negligible in few-shot XAI contexts, as FSL inherently operates with minimal shot counts. Quantitative evaluations presented in Table V confirm robust performance for both 1-shot and 5-shot settings, with these often being optimal or near-optimal. For ensuring fair comparison with baselines like SINEX(C), which are typically evaluated in 1-shot, and to represent a challenging FSL scenario, our primary quantitative results often use 1-shot, though 5-shot is also extensively validated.

V. CONCLUSION

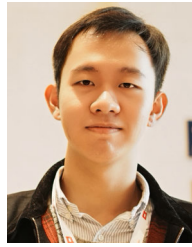
This article introduced C3A(-E), a novel XAI framework for FSL that leverages contrastive local descriptors through NBNN principles and Fisher score-based attribution. Our approach overcomes limitations in existing methods by providing comprehensive explanations through an image-to-multiclass measure. Experiments across four FSL benchmarks demonstrate C3A's effectiveness in explaining FSL backbones from both query and episodic perspectives. Moreover, a comprehensive runtime analysis in Section S.E confirms the practical feasibility of our method. Despite these advancements, we acknowledge a limitation regarding the construction of the intraclass pool: inevitably, included background descriptors in the support set may introduce noise, particularly in scenes with extreme background clutter. Future work could address this by integrating unsupervised saliency maps or weak segmentation masks to prefilter the descriptor pools, thereby further purifying the attribution sources. This work advances trustworthy FSL systems for data-scarce domains, filling the blank for posthoc FSL XAI framework.

REFERENCES

- [1] M. Hu, H. Chang, Z. Guo, B. Ma, S. Shan, and X. Chen, "Understanding few-shot learning: Measuring task relatedness and adaptation difficulty via attributes," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 36, 2023, pp. 19397–19409.
- [2] L. Steinar and B. Luca, "On episodes, prototypical networks, and few-shot learning," in *Proc. Adv. Neural Inform. Process. Syst.*, vol. 34, 2021, pp. 24581–24592.
- [3] T. Leng, Y. Zhang, K. Han, and X. Xie, "Self-sampling meta SAM: Enhancing few-shot medical image segmentation with meta-learning," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2024, pp. 7925–7935.
- [4] S. Huang, T. Xu, N. Shen, F. Mu, and J. Li, "Rethinking few-shot medical segmentation: A vector quantization view," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 3072–3081.
- [5] D. Huynh, E. Fons, and S. Vyetrenko, "Few-shot learnable augmentation for financial time series prediction under distribution shifts," in *Proc. Adv. Neural Inform. Process. Syst.*, vol. 3554, 2022, pp. 1–14.
- [6] J. Yang, X. Guo, Y. Li, F. Marinello, S. Ercisli, and Z. Zhang, "A survey of few-shot learning in smart agriculture: Developments, applications, and challenges," *Plant Methods*, vol. 18, no. 1, pp. 1–12, Dec. 2022.
- [7] A. Fedele, "Explain and interpret few-shot learning," in *Proc. World Conf. Explainable Artif. Intell.*, vol. 33, 2023, pp. 2568–2573.
- [8] W. Li et al., "LibFewShot: A comprehensive library for few-shot learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 12, pp. 14938–14955, Dec. 2023.

- [9] B. M. Lake, R. Salakhutdinov, J. Gross, and J. B. Tenenbaum, "One shot learning of simple visual concepts," *Cogn. Sci.*, vol. 33, Jul. 2011.
- [10] W. Qiyang, W. Ruiping, and C. Xilin, "Interpretable object recognition by semantic prototype analysis," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2024, pp. 789–798.
- [11] L. Fengyin, L. Mingkan, L. Da, H. Timothy, S. Yi-Zhe, and Q. Yonggang, "Zero-shot everything sketch-based image retrieval, and in explainable style," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 23349–23358.
- [12] G. Carloni, E. Pachetti, and S. Colantonio, "Causality-driven one-shot learning for prostate cancer grading from MRI," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2023, pp. 2616–2624.
- [13] P. Jingyu, L. Qi, Y. Linan, Z. Zaixi, Z. Kai, and S. Yunhao, "Towards few-shot self-explaining graph neural networks," *Mach. Learn. Knowl. Discov. Database*, vol. 14946, pp. 109–126, Aug. 2024.
- [14] M. R. Zarei and M. Komeili, "Interpretable few-shot learning with online attribute selection," *Neurocomputing*, vol. 614, Jan. 2025, Art. no. 128755.
- [15] B. Wang, L. Li, M. Verma, Y. Nakashima, R. Kawasaki, and H. Nagahara, "MTUNet: Few-shot image classification with visual explanations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2021, pp. 2294–2298.
- [16] A. Fedele, R. Guidotti, and D. Pedreschi, "Explaining Siamese networks in few-shot learning," *Mach. Learn.*, vol. 113, no. 10, pp. 7723–7760, Oct. 2024.
- [17] B. Zhang et al., "MetaDT: Meta decision tree with class hierarchy for interpretable few-shot learning," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 6, pp. 2826–2838, Jun. 2023.
- [18] Z. Ze, W. Pan, Z. Tianqi, L. Minyao, and Z. Xiaokang, "Trustworthy generative few-shot learning-based intrusion detection method in Internet of Things," *IEEE Trans. Consum. Electron.*, vol. 71, no. 1, pp. 1992–2002, Feb. 2025.
- [19] D. Bhusal et al., "SoK: Modeling explainability in security analytics for interpretability, trustworthiness, and usability," in *Proc. 18th Int. Conf. Availability, Rel. Secur.*, Aug. 2023, pp. 1–12.
- [20] M. Perlmutter and S. Krenning, "The impact of example-based XAI on trust in highly-technical populations," *Proc. Hum. Factors Ergonom. Soc. Annu. Meeting*, vol. 67, no. 1, pp. 1386–1392, Sep. 2023.
- [21] F. Mumuni and A. Mumuni, "Improving deep learning with prior knowledge and cognitive models: A survey on enhancing explainability, adversarial robustness and zero-shot learning," *Cognit. Syst. Res.*, vol. 84, Mar. 2024, Art. no. 101188.
- [22] K. Philip, J. Daniel, R. Arne, and D. Rüdiger, "Interactive teaching for fine-granular few-shot object recognition using vision transformers," in *Proc. IEEE Int. Conf. Image Process.*, Jul. 2024, pp. 535–541.
- [23] R. Ying, D. Bourgeois, J. You, M. Zitnik, and J. Leskovec, "Gnnexplainer: Generating explanations for graph neural networks," in *Proc. Adv. Neural Inform. Process. Syst.*, 2019, pp. 9244–9255.
- [24] V. J. Jeya, J. Noor, Y.-H. Cheng, L. Garcia, and M. Srivastava, "How can I explain this to you? An empirical study of deep neural network explanation methods," in *Proc. Adv. Neural Inform. Process. Syst.*, vol. 33, 2020, pp. 4211–4222.
- [25] H. Samuel and D. Gregory, "ProtoShotXAI: Using prototypical few-shot architecture for explainable AI," *J. Mach. Learn. Res.*, vol. 24, no. 1, pp. 15453–15501, 2023.
- [26] J. Snell, K. Swersky, and R. Zemel, "Prototypical networks for few-shot learning," in *Proc. 31st Int. Conf. Neural Inf. Process. Syst. (NIPS)*. Red Hook, NY, USA: Curran Associates, 2017, pp. 4080–4090.
- [27] W.-Y. Chen, Y.-C. Liu, Z. Kira, Y.-C. F. Wang, and J.-B. Huang, "A closer look at few-shot classification," in *Proc. Int. Conf. Learn. Represent.*, 2019, pp. 1–17.
- [28] W. Peizheng, Z. Qifei, Z. Jie, L. Gang, and W. Chao, "Feature transformation for few-shot learning," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Jun. 2024, pp. 1–8.
- [29] Z. Li, Z. Hu, W. Luo, and X. Hu, "SaberNet: Self-attention based effective relation network for few-shot learning," *Pattern Recognit.*, vol. 133, Jan. 2023, Art. no. 109024.
- [30] N. Catalano, A. Maranelli, A. Chiatti, and M. Matteucci, "More than the sum of its parts: Ensembling backbone networks for few-shot segmentation," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Jun. 2024, pp. 1–8.
- [31] S. Wang, J. Lu, H. Xu, Y. Hao, and X. He, "Feature mixture on pre-trained model for few-shot learning," *IEEE Trans. Image Process.*, vol. 33, pp. 4104–4115, 2024.
- [32] W. Li, L. Wang, J. Xu, J. Huo, Y. Gao, and J. Luo, "Revisiting local descriptor based image-to-class measure for few-shot learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 7253–7260.
- [33] Q. Song, S. Zhou, and D. Chen, "Learning more discriminative local descriptors with parameter-free weighted attention for few-shot learning," *Mach. Vis. Appl.*, vol. 35, no. 4, pp. 1–12, Jul. 2024.
- [34] J. Huang, F. Chen, K. Wang, L. Lin, and D. Zhang, "Enhancing prototypical few-shot learning by leveraging the local-level strategy," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2022, pp. 1660–1664.
- [35] Y. Liu, Y. Bai, H. Zhang, and X. Che, "Few-shot image classification based on cross-dimensional interactive attention," in *Proc. 5th Int. Conf. Artif. Intell. Pattern Recognit.*, Sep. 2022, pp. 810–815.
- [36] C. Finn, P. Abbeel, and S. Levine, "Model-agnostic meta-learning for fast adaptation of deep networks," in *Proc. Int. Conf. Mach. Learn.*, vol. 70, 2018, pp. 1126–1135.
- [37] Z. Li, F. Zhou, F. Chen, and H. Li, "Meta-SGD: Learning to learn quickly for few-shot learning," 2017, *arXiv:1707.09835*.
- [38] S. Ravi and H. Larochelle, "Optimization of image embeddings for few shot learning," in *Proc. Int. Conf. Learn. Represent.*, 2018, pp. 236–242.
- [39] W. Zhonghua, S. Xiangxi, L. Guosheng, and C. Jianfei, "Learning meta-class memory for few-shot semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 517–526.
- [40] R. Agrawal, "Adaptive few-shot learning (AFSL): Tackling data scarcity with stability, robustness, and versatility," 2025, *arXiv:2501.13479*.
- [41] G. Koch, R. Zemel, and R. Salakhutdinov, "Siamese neural networks for one-shot image recognition," in *Proc. Int. Conf. Mach. Learn. Worksh.*, vol. 2, M. Ranzato, A. Beygelzimer, Y. Dauphin, P. Liang, and J. W. Vaughan, Eds., 2015.
- [42] V. Oriol, B. Charles, L. Timothy, K. Koray, and W. Daan, "Matching networks for one shot learning," in *Proc. 30th Int. Conf. Neural Inf. Process. Syst. (NIPS)*, Dec. 2016, pp. 3637–3645.
- [43] G. Lin, Y. Xu, H. Lai, and J. Yin, "Revisiting few-shot learning from a causal perspective," *IEEE Trans. Knowl. Data Eng.*, vol. 36, no. 11, pp. 6908–6919, Nov. 2024.
- [44] J. Wen, Y. Liu, C. Huang, C. Liu, Y. Xu, and X. Cao, "Causal interventional prompt tuning for few-shot out-of-distribution generalization," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 48, no. 2, pp. 1978–1991, Feb. 2026.
- [45] J. Li et al., "Few-shot SAR target recognition via causal inference and deep metric learning," *IEEE Access*, vol. 13, pp. 124988–125002, 2025.
- [46] W. Jiang et al., "Causal disentanglement and cross-modal alignment for enhanced few-shot learning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2025.
- [47] N. Mishra, M. Rohaninejad, X. Chen, and P. Abbeel, "A simple neural attentive meta-learner," in *Proc. Int. Conf. Learn. Represent.*, 2018.
- [48] B. M. Lake, R. Salakhutdinov, and J. B. Tenenbaum, "Human-level concept learning through probabilistic program induction," *Science*, vol. 350, no. 6266, pp. 1332–1338, Dec. 2015.
- [49] A. Chattopadhyay, A. Sarkar, P. Howlader, and V. N. Balasubramanian, "Grad-CAM++: Generalized gradient-based visual explanations for deep convolutional networks," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Mar. 2018, pp. 839–847.
- [50] M. T. Ribeiro, S. Singh, and C. Guestrin, "Why should I trust you?: Explaining the predictions of any classifier," in *Proc. ACM Int. Conf. Knowl. Discov. Data Min.*, 2016, pp. 1135–1144.
- [51] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Proc. Adv. Neural Inform. Process. Syst.*, vol. 30, 2017, pp. 4768–4777.
- [52] V. Petsiuk, A. Das, and K. Saenko, "Rise: Randomized input sampling for explanation of black-box models," in *Proc. Br. Mach. Vis. Conf.*, 2018, p. 151.
- [53] K. Simonyan, A. Vedaldi, and A. Zisserman, "Deep inside convolutional networks: Visualising image classification models and saliency maps," in *Proc. Int. Conf. Learn. Represent. Worksh.*, 2014.
- [54] R. M. Tulio, S. Sameer, and G. Carlos, "Anchors: High-precision model-agnostic explanations," in *Proc. AAAI*, vol. 32, 2018, pp. 1527–1535.
- [55] A. Julius, G. Justin, M. Michael, G. Ian, H. Moritz, and K. Been, "Sanity checks for saliency maps," in *Adv. Neural Inform. Process. Syst.*, 2018, pp. 9525–9536.
- [56] L. Béthune, T. Boissin, T. Fel, F. Mamalet, and M. Serrurier, "On the explainable properties of 1-lipschitz neural networks: An optimal transport perspective," in *Proc. Adv. Neural Inf. Process. Syst.*, 36, 2023, pp. 54645–54682.
- [57] A. M. Salih et al., "A perspective on explainable artificial intelligence methods: SHAP and LIME," *Adv. Intell. Syst.*, vol. 7, no. 1, Jan. 2025, Art. no. 2400304.

- [58] D. Slack, S. Hilgard, E. Jia, S. Singh, and H. Lakkaraju, "Fooling LIME and SHAP: Adversarial attacks on post hoc explanation methods," in *Proc. AAAI/ACM Conf. AI, Ethics, Soc.*, Feb. 2020, pp. 180–186.
- [59] A. Barredo Arrieta et al., "Explainable artificial intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI," *Inf. Fusion*, vol. 58, pp. 82–115, Jun. 2020.
- [60] E. Jo, D. Kang, S. Kim, Y. Choi, and M. Cho, "Few-shot pattern detection via template matching and regression," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2025.
- [61] J. Donnelly, Z. Guo, A. J. Barnett, H. McTavish, C. Chen, and C. Rudin, "Rashomon sets for prototypical-part networks: Editing interpretable models in real-time," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 4528–4538.
- [62] Y. Chen, Y. Li, H. Mao, X. Chai, and L. Jiao, "A novel deep nearest neighbor neural network for few-shot remote sensing image scene classification," *Remote Sens.*, vol. 15, no. 3, p. 666, Jan. 2023.
- [63] S. McCann and D. G. Lowe, "Local naive Bayes nearest neighbor for image classification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2012, pp. 3650–3656.
- [64] K. V. Mardia, "Fisher's pioneering work on discriminant analysis and its impact on artificial intelligence," *J. Multivariate Anal.*, vol. 203, Sep. 2024, Art. no. 105341.
- [65] O. Russakovsky et al., "ImageNet large scale visual recognition challenge," *Int. J. Comput. Vis.*, vol. 115, no. 3, pp. 211–252, Dec. 2015.
- [66] A. Khosla, N. Jayadevaprakash, B. Yao, and L. Fei-Fei, "Novel dataset for fine-grained image categorization: Stanford dogs," in *Proc. IEEE Conf. Comput. Vis. Pattern Recog. Worksh.*, Jun. 2011.
- [67] K. Jonathan, S. Michael, D. Jia, and F.-F. Li, "3D object representations for fine-grained categorization," in *Proc. IEEE Int. Conf. Comput. Vis. Workshops*, Dec. 2013, pp. 554–561.
- [68] C. Wah, S. Branson, P. Welinder, P. Perona, and S. J. Belongie, "The caltech-UCSD birds-200-2011 dataset," California Inst. Technol., Pasadena, CA, USA, Tech. Rep. CNS-TR-2010-001, 2011.
- [69] Y. Xiaomeng, L. David, H. William, and D. Mehmet, "Applying class-to-class Siamese networks to explain classifications with supportive and contrastive cases," in *Proc. Int. Conf. Case-Based Reasoning*, vol. 12311, 2020, pp. 245–260.
- [70] N. Hama, M. Mase, and A. B. Owen, "Deletion and insertion tests in regression models," *J. Mach. Learn. Res.*, vol. 24, no. 1, pp. 1–38, 2023.



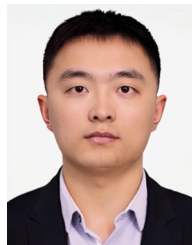
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